Learning-based spin estimation of table tennis ball with an event camera

Progress report week 1

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Summary

More work on the simulator:

- Created table tennis ball with logo in Blender (Figure 1)
- Ball spinning animation now working with spins that can be randomly generated.
- Created animation that includes ball physics. A realistic ball that can be shot and is affected by gravity. Probably overkill, just using a ball that moves in a line would be enough i think.
- Spinning ball with the creation of the events still has some bugs until i figure out how to properly set all the parameters of the simulated sensor.

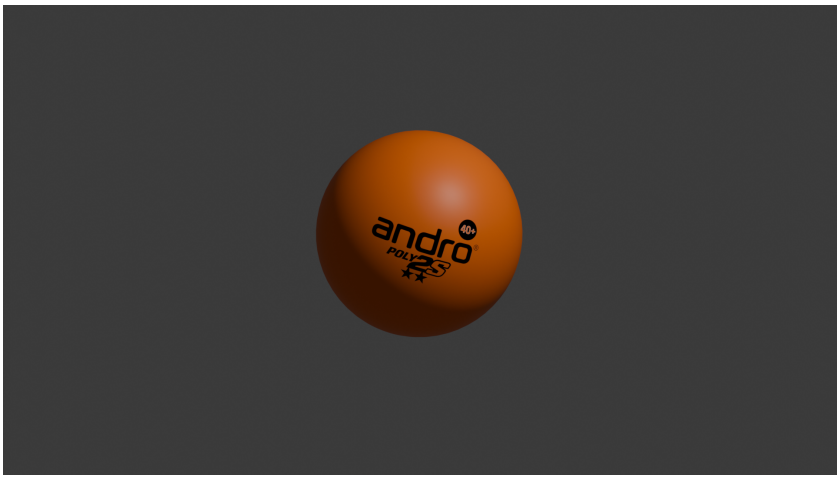

Figure 1: Table tennis ball with andro logo

New ideas

Open questions

- What are good values for max-spin? In maxs Thesis he used 0-20 rps in the simulator, but for the real data it was faster.

Next steps

- Fix bugs of the event rendering
- Align parameters with specification of real world, image-size, ball size in pixels etc.
- Implement ground truth saving (ball-position and rotation). To be able to cut the ball-region out of the image / event-stream.
- Create first dataset with simple ball rotation
- Implement ball movement
- Create dataset with moving ball

Bibliography